

Institute of Actuaries of India

Subject SP5 –Investment and Finance

July 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i)
- A higher credit rated company will have absolute advantage in borrowing at both fixed and floating rate over low rated company. A low rated company may have a comparative advantage in either fixed or floating rate.
 - A Quality Spread Differential (QSD) is the difference between the rate differential on the fixed-rate debt and the rate differential on the floating-rate debt. A positive QSD means there is comparative advantage and the swap will be beneficial to both parties by reducing overall cost of financing.

Company	Fixed rate	Floating rate	Fixed rate differential	Floating rate differential	QSD	QSD net of fee
A	9% p.a.	IBOR + 0.60%				
B	8% p.a.	IBOR – 0.25%	1.00%	0.85%	0.15%	0.10%
C	10% p.a.	IBOR + 1.55%	1.00%	0.95%	0.05%	0.00%
D	11% p.a.	IBOR + 2.45%	2.00%	1.85%	0.15%	0.10%

- Usually, the net QSD is not divided equally between both parties and the company with the lower credit rating gets the smaller share in the benefits.
- Though there is a tie between B and D to have maximum net QSD, Company A might consider entering swap with company D to get more share in benefits as company D is low rated than company B.
- However, to avail benefit of lower financing cost by swap where Company A pays fixed and receive floating rate, counterparty should have comparative advantage in borrowing at fixed rate. This is the case with Company B whereas Company D has comparative advantage in borrowing at floating rate w.r.t. Company A. Hence, Company A should enter swap with company B.

[Max 4]

- ii) Credit risk is the risk that counterparties to an agreement will be unable or unwilling to fulfill their obligations.

It can be controlled:

- by limiting the creditworthiness of the counterparties
- by limiting the total exposure to each counterparty
- using credit derivatives

[Max 2]

- iii) With a downward-sloping yield curve, the fixed rate receiver will:

- have a net outflow in the early years
- expect a net inflow in later years

The net present value of the expected cash flows should be (approximately) zero at outset. Therefore, the credit exposure is zero at outset.

Immediately after the first payment (a net outflow), the remaining payments will have a positive value to the fixed rate receiver. Consequently, there will be a small credit exposure.

As the second, third, fourth ... net outflows occur, the value of the remaining payments will increase. Consequently, the credit exposure gradually builds up.

In the middle of the life of the swap, the outstanding payments should have a large positive net present value to the fixed rate receiver. Therefore, the credit risk is large.

As the number of outstanding payments reduces their present value will fall and hence the credit exposure falls.

The credit risk reduces back to zero immediately after settlement of the final payment.

[Max 3]

- iv) The opportunity set is the set of combinations of means and variances that the investor is able to obtain by constructing portfolios containing the available securities.

A portfolio is efficient if there is no other portfolio with either a higher mean and the same or lower variance, or a lower variance and the same or higher mean. The efficient frontier is the set of efficient portfolios in E - V space.

Indifference curves join points of equal expected utility in E - V space.

The optimal portfolio is the portfolio that maximizes the investor's expected utility as a function of the mean and variance of investment returns. It is represented by a point on the efficient frontier where an indifference curve is tangential.

[Max 4]

[13 Marks]

Solution 2:

- i) Let S be the settlement price of ABC Share at end of the month;
Consider the payoffs of the Strategy A:
Short call = $30 - \max(0, S - 1000)$
Long call = $-10 + \max(0, S - 1050)$
- If $S < 1000$ then,
Payoff = $30 - 10 = 20$
 - If $S \geq 1000$ and $S < 1050$ then,
Payoff = $30 - (S - 1000) - 10 = 1020 - S$
 - If $S > 1050$ then,
Payoff = $30 - (S - 1000) - 10 + (S - 1050) = -30$
 - Strategy A is profitable when ABC share price is below 1020; breaks-even at 1020; and turns into loss making above 1020.
 - Maximum profit is 20 when $S < 1000$ and maximum loss is 30 when $S > 1050$
 - The trader might choose Strategy A if he thinks ABC share price is going to fall i.e. bearish view. Strategy A is commonly known as bearish call credit spread.
- ii) Let K is strike price and C be the call option premium at end of the month whose term to maturity was two months; $C = I_{val} + T_{val}$ i.e. intrinsic value plus time value.
Consider the payoffs of the Strategy B:

[Max 3]

Short call = $30 - \max(0, S - 1000)$

Long call = $-45 + C$

- If $S < 1000$ then I_{val} is zero and
Payoff = $30 - 45 + C = C - 15 = T_{val} - 15$
- If $S \geq 1000$ then I_{val} is $(S - 1000)$ and
Payoff = $30 - (S - 1000) - 45 + C = 30 - (S - 1000) - 45 + (S - 1000) + T_{val} = T_{val} - 15$
- Strategy B payoff is $T_{val} - 15$, irrespective of the value of S , and is profitable when $T_{val} > 15$, breaks-even when $T_{val} = 15$ and turns into loss making when $T_{val} < 15$.
- Maximum profit (or minimum loss) occurs when T_{val} is maximum; this will happen when $S=K=1000$ since we know that time value of an option is maximum for ATM (At-The-Money) options.
- Assuming all factors affecting option price (i.e. T_{val}) like volatility ' σ ' and interest rate ' r ' are not changed during the month, we can estimate maximum profit to be close to 15 (= [Call option # 2] - 15 = 30 - 15). If volatility increases (decreases) then payoff (= $T_{val} - 15$) will increase (decrease).
- Maximum loss (or minimum profit) occurs when T_{val} is minimum; this will happen when S is far away from $K=1000$ on either side i.e. deep ITM (In-The-Money) or deep OTM (Out of-The-Money). Since T_{val} cannot be negative it can go as low as zero, hence maximum loss would be 15 (= $0 - 15$).
- T_{val} decreases as S moves away from K and depends on difference between S and K . Higher absolute difference between S and K leads to lower T_{val} .
- Strategy B will have two break-even points (BEPs) when $T_{val} = 15$, one when $S < 1000$ and another when $S > 1000$.

Again assuming all factors affecting option price have not changed:

- Consider $S = 952 \left(= K \left(\frac{st}{K3} \right) = 1000 \left(\frac{1000}{1050} \right) \right)$ where premium C (= T_{val} as it is OTM) is estimated to be 9.52 $\left(= S \left(\frac{C3}{st} \right) = 952 \left(\frac{10}{1000} \right) \right)$.
- At $S = 952$, $T_{val} = 9.52$ and to increase T_{val} to 15, S should move closer to K (=1000). We can estimate lower BEP to be above 952.
- Consider $S = 1081 \left(= K \left(\frac{st}{K1} \right) = 1000 \left(\frac{1000}{925} \right) \right)$ where premium C (= $81 + T_{val}$ as it is ITM) is estimated to be 91.90 $\left(= S \left(\frac{C1}{st} \right) = 1081 \left(\frac{85}{1000} \right) \right)$ giving $T_{val} = 10.90$
- At $S = 1081$, $T_{val} = 10.90$ and to increase T_{val} to 15, S should move closer to K (=1000). We can estimate upper BEP to be below 1081.
- The trader might choose Strategy B if he thinks ABC share price is going to stay close to current price where profit will be maximum.

- The trader has no directional view i.e. neutral view since he gets profit even if share price rises (falls) slightly up to upper (lower) BEP.
Strategy B is commonly known as Neutral call calendar spread.

[Max 6]

- iii)
 - Strategy C is combination of Strategy A and Strategy B.
 - Trader's view on ABC share price is combination of bearish and neutral so net view would be mild bearish.

[Max 2]

- iv) Impact of new information was expected > 5% but only 2% realized on the day of news. With this bad news, we may expect further down move.

Convert strategy C in to two lots of strategy A that has bearish view. Sell call with a strike price of 1000 having two months to expiry and buy one more call with a strike price of 1050 having one month to expiry.

[Max 2]

[13 Marks]**Solution 3:**

- i) In life insurance business, the customer is paying the insurance company an annual premium to ensure that on his death a certain sum (sum assured) is paid to the nominees or beneficiaries, typically, spouse or children.

The insurance company is receiving this premium from numerous other individual customers annually, say from hundreds of thousands of customers. However, very small fraction (below 1%) of the customers is going to die in any given year.

The profit generated from a single sale is realized over a long period of time ranging from 1 year to 100 years depending on the type of policy sold to the customer. Profits from a policy sold accrue over a period and majority of expenses are booked immediately.

The reported earnings do not show a true picture due to

- upfront recognition of high acquisition costs, and
- need to set up reserves for future policyholder liabilities in the initial period itself that are valued with prudent margins.

Book value of a life insurance company is

- understated to the extent of margins built in the reserves for future policyholder liabilities, and
- based on amortized cost of assets rather than market value of assets.

Hence, Generally Accepted Accounting Principles (GAAP) does not reveal the true economics of the life insurance business. Therefore, the use of traditional ratios like Price-to-Earnings or Price-to-Book Value does not make sense.

In addition to the commonly used assumptions, by any company in general like sales, expenses etc. for estimation of future earnings, lot of probabilistic assumptions are required like:

- mortality or morbidity rates
- persistency related policy lapse and revival rates

These assumptions also vary

- by sales channel used like agents, bancassurance, online sales etc;

- occupation, geographic location rural / urban, life style – diet habits, smoker or not, education etc of policyholders.

[Max 4]

- ii) We can value a life insurance company by valuing future expected cash flows from all policies of each policyholder over whole policy term/duration. The expected surplus left after paying off a benefit (death or maturity) to policyholder is the profit to the company.

Thus, each policy has an embedded value which is positive if the insurance premium is priced properly and future evolves more or less as per assumptions made in the future projections.

Moreover, there is additional value from selling additional policies to the same customer and value to the company from getting new customers.

Embedded Value (EV) and Price to EV (PEV)

EV of the company is the sum of the present value of the insurer's future profits from existing policies and its net worth (total assets minus the liabilities).

EV incorporates assets at market consistent values; takes into account the margins in the policyholder liabilities and eliminates the distortion while considering one year's profits by allowing for long-term nature of insurance contracts.

EV does not account for value of future new business value. The market capitalization of a life insurer as PEV i.e. X times EV incorporates the value of new business the company is likely to generate in the future.

Return on EV (REV)

Unlocking of the value in insurance companies takes a long time, so investors should consider the movement of EV

REV is the proportion of profits a company is able to release each year out of the embedded value it has locked in. Thus, it represents the efficiency with which the company is running and validates the assumptions the company has built into its EV.

VNB and VNB margin

VNB is often termed embedded value at the point of policy sale – it represents the present value of future profits from policies issued in a specific period. It is the amount of additional EV a company generates by writing new business in a financial year.

An alternative way to value life insurer to PEV (X times EV) is PAV i.e. X times Appraisal Value (AV). $AV = EV + n \times VNB$ where n can be considered as the number of years for which the insurer expects to add same VNB as last year.

VNB margin is the profit margin of new business. This is the additional VNB generated for every Rs 100 premium of new business written by a life insurer. Higher the margin, the more favorable it is for the company and its shareholders.

Other metrics for comparing life insurers are market share, product mix, persistency ratio etc.

- **Market share** is measured by new business premium collections for a period.
 $APE \text{ (annualized premium equivalent)} = \text{regular annualized premium} + 10 \text{ percent of the new single premiums.}$

- **Product mix:** Different lines of business have different VNB margins. Generally, protection plans (primarily term plans) have higher VNB margins, followed by savings plans (primarily ULIPs).
- **Persistency ratio:** A higher persistency ratio is a favorable metric – typically, 13th and 61st month persistency ratios are considered. It indicates the insurer's customer retention history.

[Max 5]

- iii) The central bank may further increase base interest rates gradually until the economy reaches target inflation level.

Inflation can be controlled by reducing aggregate demand. However, cutting aggregate demand may result in a lower level of output and higher unemployment.

Central bank can reduce aggregate demand by lowering money supply by

- selling of securities in OMO (open market operations)
- increasing cash reserve ratios and
- increasing statutory reserve ratios

Government can reduce aggregate demand by:

- reducing government spending,
- increasing direct taxes (e.g. income tax)

Cost push inflation due to rising oil and coal prices can be controlled:

- by reducing dependency on crude oil and coal used as source of energy
- government should encourage use of electric vehicles, alternative power generation sources like solar cells / windmills etc.
- reducing indirect taxes (e.g. sales tax, customs and excise duties),

Additional measures may be used to reduce expectations of inflation:

- by the government's direct controls on prices or incomes / wages
- by Central bank's forward guidance on its stance to control inflation.

[Max 4]

- iv) Environmental issues at individual level:

Avoid investing in fossil fuel industry or any other one with high carbon emissions

- directly through stocks / bonds
- indirectly through mutual funds / bank loans from your deposits

Eat more plant based food and less animal foods

- more plants for fruits, vegetables, legumes, nuts, and cereals increases natural absorption of greenhouse gases
- fertilizer industry causes pollution; regenerative / organic farming is a solution
- raising livestock (meat / dairy industry) produces large quantity of methane
- Animal husbandry is one of the leading cause for deforestation that reduces natural absorption of greenhouse gases

Housing choices:

- Cement industry and burning bricks causes lot of pollution

- Choose traditional houses built with wood / bamboo / mud or clay and stones with plant based roof tops ;like wooden planks, palm leaves or certain grasses
- Explore durable house designs with stone blocks e.g. temples / pyramids

Travel eco-friendly like

- Walk / bicycle for short distances
- Public transport for long distances
- Ride sharing instead of solo car trips
- If possible work from home to avoid regular travelling
- Plan holidays near to home
- Switch to electric / hybrid vehicles
- Prefer trains to flights, if flight is necessary then choose non-stop flights as most fuel is used in take-offs and landings

To reduce wastage:

- Reuse items and don't replace if possible to repair
- Recycle items and if possible avoid buying single use and throw items

Fashion choices:

- Use organic clothing made from bamboo, cotton, hemp etc avoid synthetic materials like polyester
- Avoid synthetic nail polish/hair dyes instead use henna (mehandi)/other herbs

Use energy efficient appliances

- Use ceiling / hand / table fans instead of ACs as far as possible
- Adequate home insulation helps keeping warm air inside in winter and hot air outside in summers
- Replace older less efficient appliances with new technology efficient ones like bulbs with LED lights

Environmental issues at government level

- Install clean energy systems like wind mills / solar cell farms to replace oil/coal powered thermal plants
- Provide subsidies and tax reliefs to private players venturing in production of clean energy systems
- Promote electric vehicles powered by renewable energy sources
- Forestation of government waste lands with plantation of trees, mangroves etc
- introduce regular taxes on personal vehicles to encourage public transport

Target self-dependence at country level and at district / block level also.

- encourage local production to catering consumption demand as far as possible
- reduces transport costs and pollution from vehicles

Remove subsidies and /or increase tax or apply ban or prescribe limitations on use of

- fossil fuels to encourage renewable energy sources
- fertilizers to encourage organic farming

[Max 6]

- v)
- a) Herding
 - b) Mood based anomaly OR Calendar effect
 - c) Noise trading

- d) Cross sectional momentum
- e) Time series momentum

[Max 5]
[24 Marks]

Solution 4:

- i) A company's corporate dividend policy might be affected by the tax system in the following ways:
- The split between retained profits and dividends paid out will be influenced by the amount of tax payable for investors on both types.
 - The company would usually have a choice on how to distribute profits to shareholders, either as dividends or share buy-backs, and this will be impacted by whether the tax treatment differs for various options.
 - A system that makes dividends attractive from a tax point of view may lead to pressure on the company to pay a higher rate of dividends.
 - Governments will be likely to make the corporate tax system as neutral as possible, to balance the amount of dividends versus retained profits.
 - Governments will also make sure that the tax system does not lead to companies forming a dividend policy which is misaligned with the desirability of dividends based on the economic outlook, and there is no scope for manipulation.
- [Max 3]
- ii) The factors an investor should consider to maximise net of tax returns include the following:
- Difference in rates of tax on different components of investment return
 - Split between income and capital gains, assuming a different tax rate or allowance on these
 - Timing of tax payments
 - Amount of tax payments to be deferred, where possible to reduce the present value of tax payments
 - Split of tax between tax deducted at source or requiring subsequent payment
 - Amount of tax that be reclaimed depending on amount deducted at source
 - Amount of losses or gains that can be aggregate across investment types
 - Amount of losses or gains that can be aggregate across different time periods
 - Any tax exemptions which are possible to be applied for
 - Any global tax considerations if the investments are across multiple markets.
- [Max 4]
[7 Marks]

Solution 5: With reference to behavioural finance, the situation in Actuarial can be explained as:

- The insurance companies in the 1990s partly experienced **overconfidence**.
- As interest rates were far higher in past decades, there were assumptions made that rates would not fall to much lower levels.
- Worse still, some companies may have assumed some likelihood of lower rates, but ignored this to make products more attractive given high interest rates were more prevalent.
- Companies thus assumed that annuity rate guarantees were unlikely to ever bite or cause a financial strain.
- Insurers may have estimated the future cost of offering the guarantees to be negligible, or even zero, which is not the actual experience in recent years.
- The guarantees would thus have been offered without the companies holding sufficient reserves or provisions.
- While the reserving may have been justifiable given the economic situation before 2000, it is unlikely to be acceptable given the gradually falling interest rates.
- After 2000, when interest rates began to reduce, insurers may have experienced **anchoring and adjustment**.

- Companies may have believed that any fall in interest rates was temporary only.
- Insurers may have continued to assume the previous high rates of interest rates and investment return, assuming this to be 'normal'.
- **Confirmation bias** may have been involved, if interest rates temporarily rose back to previous high levels.
- In addition, for some insurers, if the guarantees only began to bite in a few recent years, their management may have suffered from a combination of all of the biases noted above.
- An amount of **hindsight bias** could kick in for anyone looking back at the trend over the past decades, assuming that the fall in interest rates was inevitable given the original rates were too high.

[5 Marks]

Solution 6:**i) The uses of Asset pricing models are as follows:**

- The first use of asset pricing models is to determine whether an observed asset price is 'incorrect'. [1/2]
- This tells us whether the price of the asset in the market represents a trading opportunity for a smart investor. [1/2]
- The second use of asset pricing models is to determine what the asset price should be, where it cannot be observed. [1]
- **Absolute pricing** is a type of asset pricing model which refers to exposure to fundamental sources of macroeconomic risk. [1/2]
- For example, unpredictable macroeconomic variables such as inflation, economic growth and interest rates are referred to in absolute pricing. [1/2]
- Consumption-based and general equilibrium models (such as CAPM) are examples of this approach. [1/2]
- Absolute pricing can be used to predict how prices might change if the policy or economic structures change. [1/2]
- Relative pricing considers the value of an asset given the price of some other assets. [1/2]
- Examples of relative pricing are the Black-Scholes option pricing and Arbitrage Pricing Theory. [1/2]
- In relative pricing, we use as little information about fundamental risk factors as possible. [1/2]
- In relative pricing, we also do not ask where the prices of the other assets come from. [1/2]

[Max 6]

ii) Hedging longevity risk using a bond:

- Buying the bond exposes the scheme to credit risk as there may be a default on the coupons and/or the redemption payment. [1/2]
- The credit risk is unlikely to be a problem in normal economic conditions, but it could become a material risk in the event of a credit crunch or similar events. [1/2]
- The bond is likely to be highly illiquid, as it would have been issued only to the pension scheme. [1/2]
- This could cause problems if the scheme needed to sell the bond, in case it needed additional cash flows or was concerned about the default risk. [1/2]
- It may be time-consuming and expensive to invest in the bond. [1/2]
- In particular, the scheme may need to undertake in or pay for a credit rating for the bond. [1/2]
- The scheme may need to spend time and cost to negotiate the exact terms of the bond. [1/2]
- The scheme may also need to obtain necessary approvals. [1/2]
- The price paid for the bond will need to cover the profit margin required by the scheme. [1/2]

- The price should also cover the cost of designing and issuing the bond, and taking on board the longevity risk of the scheme. So it might be expensive. [1]
- The survival experience of the pension scheme is unlikely to match exactly the life table underlying the bond. This leads to basis risk and imperfect hedging. [1]

[Max 6]

iii) The Global equity unit trust invests 75% of the fund is invested passively to track a global equities benchmark, while 25% of the fund is actively invested in a range of equities.

- This way of structuring the portfolio is known as the core / satellite approach. [1/2]
- The trust manager may believe that the shares in the global equities benchmark are all efficiently priced. [1/2]
- It will not always be possible to consistently out-perform stocks from amongst the benchmark constituents to produce excess risk adjusted returns. [1/2]
- However, the passive approach for 75% of the fund will reduce fund management and dealing costs. [1/2]
- The cost will be even lower if the approach is not to hold actual shares of the benchmark constituents, but instead track the benchmark synthetically using an index futures and treasury bills. [1/2]
- There is lower performance risk compared to other trust managers if the global equity benchmark is a widely tracked index. [1/2]
- If the trust's peer group are adopting a different management approach, for example, not tracking a benchmark or a following a less portion tracking the benchmark, they might have higher risks compensated with potential higher returns for some periods. [1]
- Tracking the benchmark also ensures a degree of diversification within the unit trust, and lowers the overall volatility of investment performance relative to the market. [1/2]
- Another reason for choosing this investment approach may be to align with the marketing literature of the fund, in case investors have been selecting the trust because of this strategy. [1/2]
- Alternatively, this kind of approach may be an explicit requirement of the investment objectives of the company, or mandated by regulatory constraints. [1/2]
- The 25% actively invested in a range of equities could also be increased, depending on the efficiency of the market for various kinds of equities. [1/2]
- The investment management skills of the trust manager to exploit any inefficiencies may also alter this approach, although the trust manager needs to consider any additional costs associated with additional active trading. [1/2]

[Max 6]

[18 Marks]

Solution 7:

i) **Risk budgeting to design investment strategy for fund that awards scholarships to young actuaries and accountants:**

- Risk budgeting is concerned with determining both the amount of investment risk an investor is prepared to take, and also where it is most efficient to take that risk in order to maximise investment return. [1/2]

- This enables the investor to maximise total expected returns for a given level of investment risk. [1/2]
- Initially, it would be necessary to define a 'feasible set' of asset classes that could be included in the portfolio. [1/2]
- It is necessary to obtain expected returns, volatility and covariance data on each feasible investment category. [1/2]
- Next, an asset-liability modelling process is used to determine a suitable asset allocation to meet liabilities with an acceptable degree of certainty ('matching portfolio'). [1/2]
- The next step is then to use Value at Risk techniques to determine the overall investment risk (the 'risk budget') that the fund is prepared to take in order to achieve higher returns. [1/2]
- The risk budget will require consideration based on the risk attitude of sponsors for the scholarship fund. [1/2]
- Once the sponsors of the scholarship have determined the above, there needs to be a decision on how to allocate the risk between strategic risk (risk relative to liabilities) versus active risk (extent to which individual managers are allowed to deviate from the benchmark portfolio). [1]
- There also needs to be an allocation of the available active risk across the component portfolios and between the individual managers. [1]
- The risk allocations need to allow for the extent to which the sponsors believe that active management is able to add value. [1/2]
- Assumed correlations between strategic and active risk also needs to be considered. [1/2]
- Assumed correlations between the active risks of the individual fund managers also needs to be considered. [1/2]
- After determining and implementing the optimal asset allocation, exposures will need to be monitored subsequently both in terms of increases and decreases in exposures, and in terms of changes in volatility and correlation data. [1]
- Rebalancing should be performed in response to short-term changes in the volatility and correlation of assets. [1/2]
- As the fund is for both young accountants and actuaries, it should be considered whether there is a separate risk budget for both. This would be determined by analysing the underlying data, and observing any significant differences, such as number of accountants eligible versus actuaries each year. [1]

[Max 8]

ii) Calculations

We first need to calculate the β 's of the two investment trusts using:

$$\beta_i = \text{Cov}(R_i, R_m) / V_m$$

For Trust A, $\beta = 0.33 * 0.55 * 0.15 / (0.15 * 0.15) = 1.21$ [1/2]

For Trust B, $\beta = 0.87 * 0.25 * 0.15 / (0.15 * 0.15) = 1.45$ [1/2]

Investment Trust A

Treynor measure = $(0.04 - 0.01) / 1.21 = 0.025$ [1/2]

Sharpe measure = $(0.04 - 0.01) / 0.55 = 0.055$ [1/2]

Jensen measure = $0.04 - (0.01 + 1.21 * (0.02 - 0.01)) = 0.018$ [1/2]

Pre-specified SD = $0.04 - (0.01 + (0.02 - 0.01)/0.15 * 0.55) = -0.007$. [1/2]

<Full marks awarded in case of differences due to rounding>

Investment Trust B

Treynor measure = $(0.05 - 0.01) / 1.45 = 0.028$ [1/2]

Sharpe measure = $(0.05 - 0.01) / 0.25 = 0.16$. [1/2]

Jensen measure = $0.05 - (0.01 + 1.45 * (0.02 - 0.01)) = 0.026$ [1/2]

Pre-specified SD = $0.05 - (0.01 + (0.02 - 0.01)/0.15 * 0.25) = 0.023$ [1/2]

[Max 5]

iii) Choice of Investment Trust

- Investment Trust B slightly outperforms Trust A on all four measures. [1/2]
- Intuitively, this makes sense at a high level, as the standard deviation for Trust A is relatively large, compared with the benchmark and Trust B. [1/2]
- In particular, the two measures based on standard deviation of return (Sharpe and pre-specified SD) show a huge advantage for Trust B. [1/2]
- The two measures based on systematic risk (Treynor and Jensen) give much closer results for both trusts. [1/2]
- If the investment trust(s) are held as the sole asset of the pension fund then it is appropriate to consider the total risk. Trust B is then preferred given the far lower standard deviation. [1]
- If the investment trust(s) are held as a small part of the pension fund's investments the high specific risk of Investment Trust A will be diversified away. [1/2]
- In case the volatility of Trust A can be diversified away, then there is not much difference between the two trusts on the measures for systematic risk, so more information will be required to reach a decision. [1]
- The data is only from the last five years, and there is no guarantee that future performance will reflect the past experience. [1/2]
- For example, the investment strategy for the trusts may change. [1/2]
- Tax allowance should be made in the data, as it is not clear if this has been considered. [1/2]
- Investment expenses allowance should be made in the data, as it is not clear if this has been considered. [1/2]
- Matching of the trust assets to any of the fund's liabilities needs to be considered. [1/2]
- The Treynor and Jensen measures rest on the validity of the capital asset pricing model. [1]

[Max 7]

[20 Marks]
